Experiment - 6

Error detection & Error correction

Aim:

Write a program to implement error detection and correction using hamming code concept make a test run to input data stream and verify error correction feature.

Sender program:

Input to sender file should be a text of any length program should convert the text to binary.

Apply hamming code concept on the binary data and add redundant bits to it

Save in a file called channel

Code (sender.py)

```
data = list (map(int, input ("Enter data bits : ").strip ())
m = len (data)
r = 0
while (2**r) <(m+r+1)
    r+=1
n = m+r
hamming = [0] * (n+1)
j = 0
for (i in range (1, n+1);
    if (i (i-1)) != 0;
        hamming [i] = data [j]
        j+=1
for i in range (r);
    pos = 2**p
    Parity = 0
for i in range (1, r+1);
    if i < pos;
```

```
            parity < hamming[i]
        hamming[pos] = parity
    code = hamming[1:]
    print("Data bits :",data)
    print("hamming code :",code)
```

Output:

```
Enter data bits = 1011
Data bits : [1,0,1,1]
Hamming code : [0,1,1,0,0,1,1]
```

~~Recives~~

Reciever code (reciever.py)

```
recieved = list(map(int,input("Enter recieved hamming code").strip()))
n = len(recieved)
r = 0
while (2**r) <= n
    r+1 = 1
error pos = 0
for p in range(n)
    pos = 2**p
    parity = 0
for i in range(1,n+1)
    if i & pos:
        parity = recieved[i-1]
    if parity != 0:
        error pos = pos
    if error pos = 0;
        print("No error detected.
```

```
else:
    print("
    recieved
    print(
data[]
    for i
        if
        data
    prin
```

Output:

```
suppos
error
correct
origin
```

```
Result :
    Thu
successful
The hamm
successfull
```

```
else :
  print ("error detected at position";error-pos)
  recieved [error-pos-1]^ = 1
  print ("corrected hamming code :" ; recieved)
data []
  for i in range (1,n+1);
     if (i c[(i-1)] !, = 0;
     data .append (recieved (i-1))
     print ("original data list :", data)
```

Output :

suppose hc. 0110011 change to 0111011
error detected at position:4
corrected hamming code : [0,1,1,0,0,1,1]
original data bits : [1,0,1,1]

Result :

Thus, the hamming code algorithm was successfully implemented in python, the sender generated The hamming code from the given data bits, the reciever successfully detected and corrected a single bit errors.